Venya Chowdhery

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Dubai, UAE | UAE Golden Visa Holder

EDUCATION

Birla Institute of Technology and Science, Pilani – Dubai Campus Sep 2022 – Jul 2026
B.E. Computer Science, Minor in Data Science | CGPA: 8.5 / 10

The Indian High School, Dubai Apr 2018 – Jun 2022
Senior Secondary (CBSE) | Class XII: 91.5% | Class X: 93.6%

WORK EXPERIENCE

NEXA Digital Marketing | *Artificial Intelligence Intern* Jul 2026 – Present

- Built and deployed customer-facing **AI assistants (LLM-powered chatbots and voicebots)** on **WhatsApp** for clients across multiple industries, including **Rove Hotels**, **Nest Hotels**, and **Marjan**, reaching **44,600+ unique users**.
- Delivered **client-facing financial dashboards** (HTML, JavaScript) with an embedded **AI chatbot**, giving clients self-serve, natural-language access to performance data.
- Built a full-stack **account intelligence hub** (Next.js, TypeScript, PostgreSQL/Prisma, **Claude API**) that uses **LLMs** to auto-extract action items from **Slack, Gmail, and Fireflies** for human review, with a per-account **client dashboard** and automated **security testing** to protect user data and prevent unauthorized access.
- Engineered an **end-to-end agentic n8n + LLM automation pipeline** that turns client ad briefs into live Meta ad campaigns via client approval, plus a post-launch **reporting layer** with 24-hour performance insights and 48-hour **scale/adjust/kill recommendations**.
- Redesigned the **expense reimbursement process** by automating submission, approval, and HR handoff across Slack, Sheets, and Drive, cutting processing time by about **70%**.

Tenderd | *Artificial Intelligence Intern* Jan 2026 – Jul 2026

- Built an **end-to-end ML pipeline** in **Python** on telemetry sensor data, covering data extraction, EDA, and feature engineering, implementing **Isolation Forest** and **Autoencoder (deep learning)** models to detect **sensor malfunctions** and equipment deviations across diverse fleets.
- Engineered **rule-based detection systems** for loose sensors and data exhaustion events to identify device connectivity failures and operational anomalies, enabling **proactive equipment maintenance**.
- Automated PPE detection labelling by integrating a custom **YOLOv11-nano object detection model** into Label Studio's ML backend, reducing manual annotation overhead and enabling rapid dataset iteration for **computer vision** workflows.
- Validated anomaly flags with **engineering and operations teams** through QA workflows before fleet-wide deployment.

Oasis Investment, Al Shirawi Group | *Digital Solutions Intern* Jun 2024 – Aug 2024

- Delivered **GUI and UX improvements** for an Angular-based **Supplier Registration Web Portal**, improving form usability.
- Built **intern-tracking and progress-reporting web pages** for HR, supporting data-driven tracking of program milestones.
- Authored a **product PR/FAQ** for ConferEase, a proposed conference-room booking app, covering target users, value proposition, and feature roadmap.

TECHNICAL SKILLS

- **GenAI & Agents:** RAG, LangChain, LangGraph, AI Agents, Multi-Agent Systems, Tool Calling, ChromaDB, Embeddings, Vector Search
- **AI Engineering/ LLMOps:** LLM APIs, Prompt Engineering, Guardrails, Human-in-the-Loop, LLM Evaluation, LoRA
- **Computer Vision:** Image Classification, Object Detection (YOLO), Segmentation, ViT, Google Earth Engine
- **ML/ Programming:** Python, SQL, PyTorch, TensorFlow, Scikit-Learn, Hugging Face, CNNs, LSTM, BERT, Autoencoders, Java, C++
- **Full-Stack/ Cloud/ Tools:** FastAPI, REST APIs, Next.js, React, TypeScript, PostgreSQL, Docker, Git/GitHub, CI/CD, GCP, n8n, Power BI

PROJECTS

Agentic Research Assistant | *RAG, LLMs, LangGraph, Python*

- Built a **multi-agent Retrieval-Augmented Generation (RAG) system** that decomposes complex queries into sub-tasks, retrieves document context via **semantic vector search**, and synthesizes grounded, cited answers using LLM APIs.
- Implemented an end-to-end pipeline (ingestion, chunking, embeddings, retrieval, claim verification) orchestrated with **LangGraph**, using claim verification as a **guardrail** to reduce hallucinations and improve factual accuracy.

n8n Copilot: ML-Ranked Workflow Node Recommendation Engine | *Python, FastAPI, ML*

- Built a **Chrome extension** and **FastAPI backend** that recommends the next node while building n8n workflows, mining **494 real workflow templates** and blending transition statistics with semantic search (**Sentence-Transformers**) for real-time ranking.
- Evaluated the model on 1,547 held-out workflow edges (**54.7% Recall@5, 70.1% Recall@10**) and deployed the service on Render with CI/CD via GitHub Actions.

SummarEase: Speech-to-Text Summarization App | *Generative AI, NLP, Transformers*

- Built a transcription-and-summarization pipeline using **transformer models** (**OpenAI Whisper**, **BART**) with an integrated **LLM chatbot** for interactive Q&A on transcribed content, deployed as a **Streamlit** web app.